THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 1

Why Databricks, Why Now — Career Opportunity & Series Introduction

SECTION 01

Why Databricks, Why Now

Databricks has become one of the fastest-growing platforms in the modern data stack, and the job market reflects it directly. This section lays out the numbers behind the hook you saw in the reel.

The Growth Signal

- Databricks job postings grew over 10% year-over-year in the most recent annual period, continuing a multi-year upward trend.
- Databricks' own workforce has grown roughly 60% over the last two years, with continued double-digit year-over-year growth.
- Databricks now appears in a meaningful share of all Data Engineer job postings analyzed industry-wide, alongside Spark and Snowflake as one of the most requested platform skills.
- The Bureau of Labor Statistics projects data engineering and data science roles to be among the fastest-growing occupations through the early 2030s.

REALITY CHECK

Exact salary figures vary widely by city, company stage, and experience level. Treat any single number you see online (including in this document) as a directional signal, not a guarantee. Use it to gauge market direction, not to negotiate a specific offer.

Why This Matters For You

Databricks-related roles now span multiple titles: Data Engineer, Analytics Engineer, ML Engineer, and Platform/Lakehouse Engineer. Interviewers increasingly expect candidates to understand not just SQL and Python, but how a modern Lakehouse platform is architected end-to-end. That is exactly the gap this series is built to close.

SECTION 02

What Is Databricks, Really

Databricks is a unified data and AI platform built primarily on Apache Spark. It provides the compute, orchestration, governance, and collaboration layer that data teams use to build pipelines, run analytics, and train machine learning models.

CORE CONCEPT

Databricks does not store your data itself. Your actual data lives in your cloud object storage, such as AWS S3, Azure Data Lake Storage, or Google Cloud Storage. Databricks is the compute and management layer that sits on top of that storage, processing, governing, and refining the data without owning it.

The Problem Databricks Solves

Before platforms like Databricks, data teams typically had to choose between two imperfect options.

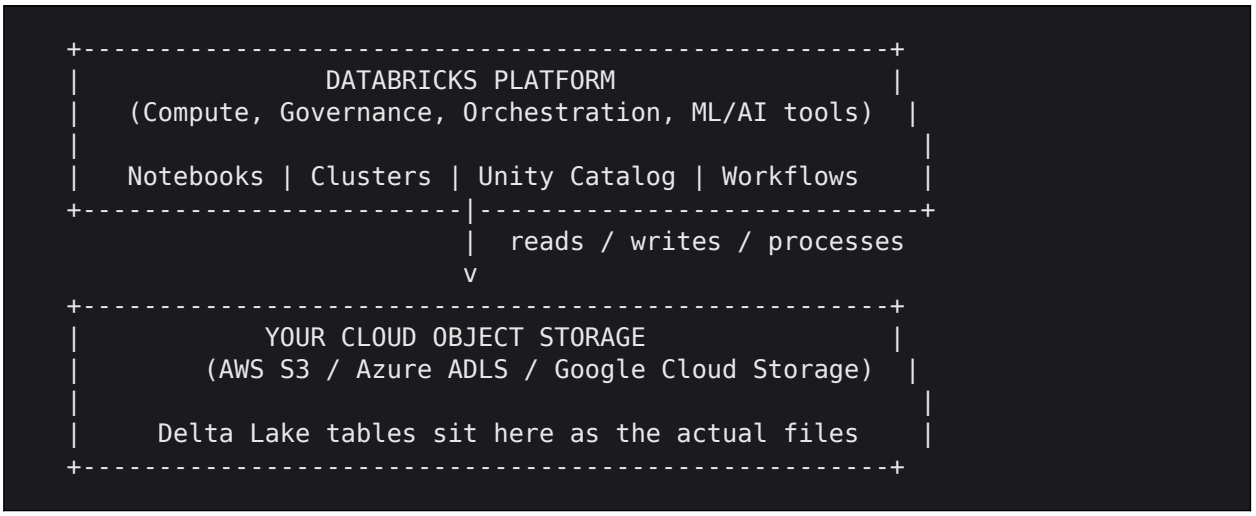
- Data Lakes: cheap, flexible storage for raw data of any format, but weak on structure, reliability, and performance.
- Data Warehouses: strong structure, reliability, and query performance, but expensive and rigid, with limited support for unstructured or semi-structured data.

Databricks popularized the Lakehouse architecture, which combines the low-cost flexible storage of a data lake with the reliability, structure, and performance features traditionally associated with a data warehouse, using Delta Lake as the underlying storage format.

SECTION 03

The Lakehouse Architecture (High Level)

A simplified view of how the pieces fit together at a conceptual level:



REMEMBER THIS FOR INTERVIEWS

A very common interview trap is assuming Databricks is a database or a storage system. The correct framing: Databricks is a unified compute and governance layer; the data itself resides in cloud object storage, typically in Delta Lake format.

SECTION 04

Data Warehouse vs Data Lake vs Lakehouse

Aspect	Data Warehouse	Data Lake	Lakehouse (Databricks)
Data Structure	Structured only	Any format (raw)	Any format, with structure enforced
Cost	Higher	Low	Low, storage-efficient

Reliability (ACID)	Strong	Weak / none	Strong, via Delta Lake
Best For	BI, reporting	Raw data storage	BI, ML, streaming, all in one
Flexibility	Low	High	High

SECTION 05

What This 30-Day Series Covers

The series is structured in five phases, moving from foundations to advanced, production-grade concepts.

Phase	Focus	Sample Topics
1. Foundations	Days 1-6	Architecture, Control vs Compute Plane, Clusters, DBU Pricing
2. Delta Lake	Days 7-13	ACID Transactions, Time Travel, Z-Ordering, Vacuum
3. Governance & Orchestration	Days 14-20	Unity Catalog, Medallion Architecture, Workflows, CI/CD
4. Streaming & Advanced Engineering	Days 21-26	Structured Streaming, Autoloader, DLT, Photon
5. Real-World & ML	Days 27-30	MLflow, Databricks vs Snowflake, End-to-End Case Study

SECTION 06

Glossary — Day 1 Terms

Term	Definition
Lakehouse	An architecture combining data lake flexibility with data warehouse reliability and structure.
Delta Lake	An open-source storage layer that brings ACID transactions and reliability to data lake files.
Cloud Object Storage	Storage services like AWS S3, Azure ADLS, or GCS where the actual data files physically reside.
Control Plane	The Databricks-managed backend (web app, job scheduling metadata) that lives in the Databricks account.
Compute Plane	Where actual data processing happens, typically inside your own cloud account.

SECTION 07

Interview Questions & Answers

Q1. Does Databricks store your data?

A. No. Databricks is a compute and governance layer. Your actual data resides in your cloud object storage, such as S3, ADLS, or GCS. Databricks processes and governs that data without owning the underlying files.

Q2. What problem does the Lakehouse architecture solve?

A. It removes the need to choose between a data lake and a data warehouse by combining the low-cost flexibility of a lake with the ACID reliability and performance of a warehouse, using Delta Lake as the storage format.

Q3. What is Delta Lake, in one line?

A. An open-source storage layer built on top of Parquet files that adds ACID transactions, schema enforcement, and time travel to data lake storage.

Q4. Is Databricks the same as Apache Spark?

A. No. Databricks is a managed platform built primarily on top of Apache Spark, adding a unified workspace, governance, orchestration, and performance optimizations like the Photon engine.

Q5. Why are companies moving to a Lakehouse model instead of maintaining separate lakes and warehouses?

A. Maintaining two systems duplicates data, increases cost, and creates consistency issues between the raw and curated layers. A Lakehouse consolidates both needs into a single governed platform.

Q6. What are the two planes in Databricks' architecture at a high level?

A. The Control Plane, managed by Databricks and containing the web application and job metadata, and the Compute Plane, where your actual data processing occurs, typically within your own cloud account.

STAY ON TRACK FOR ALL 30 DAYS

Comment the keyword **DATABRICKS** on the reel to get this document and future deep-dives sent directly to you.

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